ANESTHESIA

2026-27

CONCEPTS | CLINICALS | MCQS

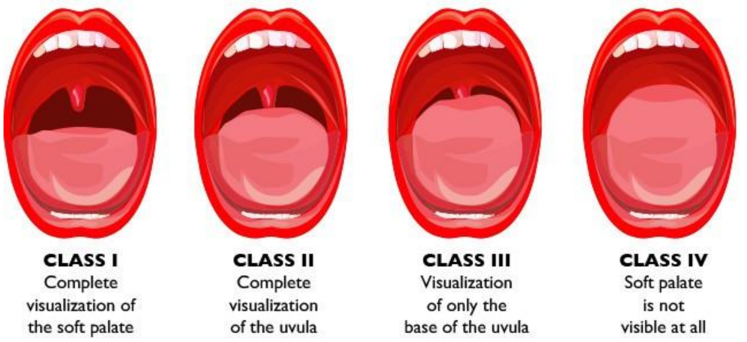
-Control, Precision, Perfection.
-Drugs, mechanisms &
clinical application simplified.
-Most expected MCQs & viva points.

DR. KEYUR GANDHI DR JITENDRA

ANESTHESIA

PAC: PRE-ANESTHESTIC CHECK UP
Airway assessment - Most important step
History of comorbidities/allergies/personal history/surgery
Investigations
Consent

MALLAMPATI CLASSIFICATION **FMGE 2020,21,23,24,26**
simple bedside test used to predict the ease of endotracheal intubation

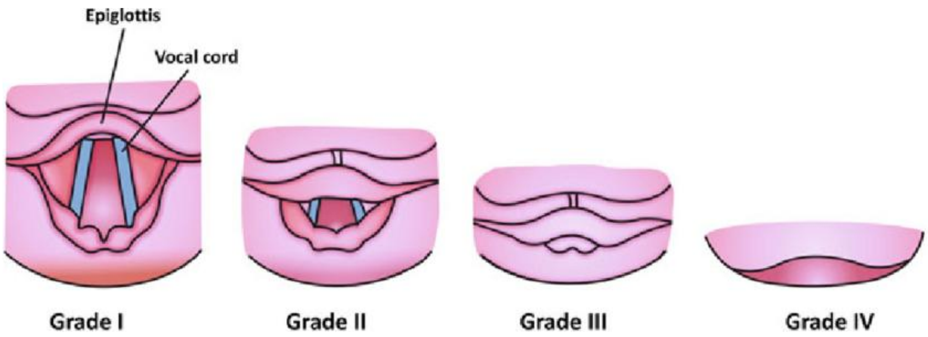


ASA CLASSIFICATION **INICET-21,23,FMGE 24**

GRADE	DEFINITION	EXAMPLES
ASA I	Normal healthy patient	**Healthy , non smoking,no or minimal alcohol use**
ASA II	**Patient with mild systemic disease**	**Mild disease only without substantive functional limitations,current smoker,social alcohol drinker,pregnancy,obesity**
ASA III	**Patient with severe systemic disease**	**Substantive functional limitations,moderate to severe. Poorly controlled DM or HTN, COPD,alcohol dependance , history of >3months of MI,STROKE,TIA or CAD.**
ASA IV	**Patient with severe systemic disease that is constant threat to life**	**Recent <3 months MI,STROKE,TIA,CAD.**
ASA V	**Moribund patient not expected to survive without operation**	**Ruptured thoracic/abdominal aneurysm,massive trauma,intracranial bleed.**
ASA VI	**Declared brain dead patient whose**	

	organs are being removed for donor purposes	

CORMACK AND LEHANE CLASSIFICATION



Most common intra-operative anaphylaxis- Antibiotics>Muscle relaxant
Pre-op antibiotic time- 30 min before incision→ CEFAZOLIN **NEET PG 2023**

COEXISTING MEDICAL ILLNESS **FMGE 25**

1. Anti HTN's - should be continued in the same dose
Exception - ACE/ARB's are discontinued
2. Anti anginal/Antithyroid/Antiepileptics are continued
3. In diabetics, OHA should be discontinued as patient is fasting during surgery & before surgery → Risk of hypoglycemia
• If required, switch to Insulin (major surgeries, 48 hours before surgery)
4. Aspirin - 75mg can be continue
5. Clopidogrel - Stopped 8 days before to the surgery
6. Warfarin - Stopped 3-5 days before to surgery
7. Heparin -Stopped 6 hours before to surgery
8. LMWH - Stopped 12 hours before to surgery
9. Ticlopidine - Stopped 12-14 days before to surgery
10. OCP,MAOi,TCA- should be stopped 3 week before to surgery
11. Herbal medications should be stopped before 6 week before surgery

LOCAL ANESTHETICS **FMGE 2020,24,INICET-21**

MOA- block Na+ channel
Main mechanism •The non-ionized form of local anaesthetic enters the nerve ଠ] The ionized form of local anaesthetic blocks the sodium channel
Classifications
1. AMINOESTERS- (1 i in name)- metabolise by plasma esterase except cocaine
Cocaine- metabolise in liver
Chlorprocaine
Procaine
Benzocaine
Proparacaine- eye surgery

2. AMINOESTERS (2 i in name)- metabolise in liver
Lignocaine
Bupivacaine
Prilocaine
Dibucaine

COCAINE **INI 2022**
Aminoester
Potent vasoconstrictor
Can not be given iv

LIGNOCAINE **FMGE 2023**
Only LA given IV
Most commonly used worldwide
Causes malignant hyperthermia

IV RA is also known as Biers block

PROCAINE
Short-acting LA
Safest LA in malignant hyperthermia

Procaine and Benzocaine metabolised to para amino benzoic acid→ causes allergy

BUPIVACAINE **NEETPG2021,INICET-21,FMGE 24**
Most commonly used LA in spinal anesthesia
Most cardiotoxic LA
toxicity-DOC: 20% intralipids

DIBUCAINE
Longest acting LA
Only used for dibucaine number test

Detects atypical pseudocholinesterase or pseudocholinesterase deficiency
Pseudocholinesterase is an enzyme responsible for the metabolism of drugs like Scoline Patient with pseudocholinesterase deficiency: Prolonged action of scoline → delayed recovery

Best topical LA for cataract surgery is Proparacaine **FMGE 2024**

PRILOCAINE
Used for surface anesthesia

Contraindicated in neonates
Can cause methhemoglobinemia (PRILOCAINE AND BENZOCAINE)

EMLA
Eutectic Mixture of LA
2.5% Lignocaine + 2.5% prilocaine
Can cause methhemoglobinemia

PARA-AMINO-BENZOIC ACID- PROCAINE + BENZOCAINE
EMLA- PRILOACINE + LIGNOCAINE **NEET PG 24**
METHHEMOGLOBINEMIA- PRILOCAINE + BENZOCAINE

SPINAL ANESTHESIA **NEETPG 2021,INI 25**

Surgery below umbilicus
location-Adult: L3-L4 or L4-L5 ,Infant-L4-L5

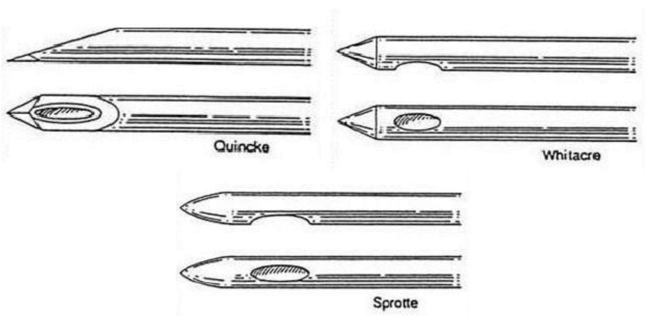
Layers pierced-
Skin → subcutaneous tissue → supraspinatous ligament → interspinous ligament → ligamentum flavum → duramater → arachnoid (last layer) → subarachnoid space

Complications of spinal anesthesia **FMGE 2025**

INTRA-OPERATIVE	POST-OPERATIVE
Hypotension- most common Rx- Phenylephrine	Acute urinary retention- most common
Shivering Rx- Pethedine	Headache
Bradycradia Rx- Atropine	6th cranial nerve palsy

EPIDURAL ANESTHESIA **FMGE 2020,21,24,INICET-21,FMGE26**

Surgery below umbilicus - long duration surgeries
location-Adult: L3-L4 or L4-L5 ,Infant: L4-L5
Layers pierced-
Skin → subcutaneous tissue → supraspinatous ligament → interspinous ligament → ligamentum flavum



Tuohy

Anterior ethmoidal nerve

GENERAL ANESTHESIA

1. Preoxygenation phase - give 3-5 min with 100 % oxygen
2. Induction phase- give IV induction agent,short acting muscle relaxant,intubate the patient
3. Maintenance phase- keep patient in unconscious state with inhalational/IV agent,long acting muscle relaxant
4. Reversal phase-stop inhalational agent, muscle relaxant antagonist→ neostigmine

(**FMGE 24,NEETPG 2020,INI 25**)

Glycopyrolate - ↓ secretions **INICET 2020**

REVERSAL AGENTS (muscle relaxant) **NEETPG 2020**

Neostigmine
Sugammadex
Pyridostigmine
Edrophonium

IV INDUCTION AGENTS **FMGE 2020,21,24,25,INI 25**

Propofol	Etomidate	Thiopentone	Ketamine
Act on GABA receptor	Act on GABA receptor	Act on GABA receptor	Act on NMDA receptor antagonist
Milky white solution 	Milky white solution	Yellow amorphous powder	-
Painful	Painful	-	-
Induction of choice- 1. Day care surgery 2. Total IV anaesthesia 3. Acute porphyria 4. LMA insertion 5. Malignant hyperthermia	anesthesia of choice for cardiac surgeries	IV induction of choice in neurosurgery	IV induction agent of choice shock,CHD, asthma

Only IV induction with anti emetic property	Produces myoclonic jerk	Ultra short acting Rapid redistribution	Only anesthesia with analgesic property
Causes hypotension and bradycardia	Side effects- adrenal suppression	Most dreadful complication- Accidental intra arterial inj. Rx- lignocaine	Side effects- ↑ secretion Dissociative anesthesia hallucinations
Reduce ICP and second best agent		CI in porphyria and asthma	CI in neurosurgery,ischemic heart disease and glaucoma

INHALATIONAL AGENTS **INICET-21,24,FMGE 2021,22,23,NEET PG 24**

SEVOFLURANE	ISOFLURANE	DESFLURANE	HALOTHANE
Sweet odour Induction Agent of choice in children	strong pungent odour		
In closed circuit it reacts with soda lime when used in low flow anesthesia	Shows coronary steal phenomenon	Fastest induction Avoid in asthma patient	Contraindicated in heart surgery Can cause Malignant Hyperthermia,hepatitis
Best inhalational agent of choice for bronchial asthma	Most commonly used for cardiac,neuro surgery	Best to use for cardiac surgery Best for liver/renal transplant surgery	Cause maximum bronchodilation,muscle relaxant
			

MAC (MINIMAL ALVEOLAR CONCENTRATION) **INI 2022**

MAC is the alveolar concentration of an inhalational anesthetic (expressed as a percentage of 1 atmosphere) at which 50% of patients do not move in response to a surgical (painful) stimulus.
It is a measure of the potency of an inhaled anesthetic.

Lower MAC = more potent drug
Higher MAC = less potent drug
Methoxyflurane>Halothane>Isoflurane>Desflurane>Xenon

BLOOD GAS PARTITION **FMGE 24**

Indicator of solubility of inhalational agent in blood
High B/G - slow onset
Low B/G - fast onset

Methoxyflurane<Halothane<Isoflurane<Desflurane<Xenon

SKELETAL MUSCLE RELAXANTS

Act on neuromuscular junction and block acetylcholine receptor

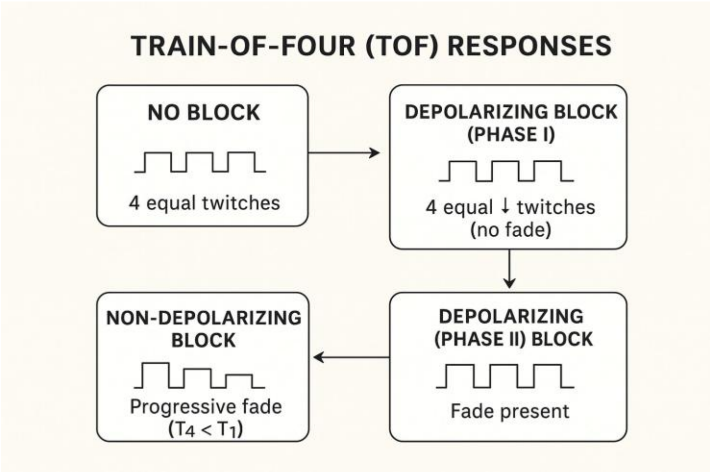
REVERSAL AGENTS
Pyridostigmine,Edrophonium,Neostigmine

MODIFIED GAMMA CYCLODEXTRIN'S
Sugammadex -Used to reverse the effects of Rocuronium and Vecuronium

SIGNS OF ADEQUATE REVERSAL

best clinical/bedside sign -Sustained head lift >5 seconds
Overall best sign- TOF ratio >0.9 (usually used to monitor the effects of muscle relaxants)

Electrodes are kept on adductor pollicis **FMGE 2022,NEETPG 2021,22**
stimulate ulnar nerve



NEET PG 2021,24,FMGE 2021,25,INI 24,NEET 2025

NDMR	DMR
Benzylisoquinolinium (curium) **Atracurium -** -metabolised by haffmann degeneration. -Indicated in renal failure, liver failure -Laudanosine is formed on long-term infusion -Histamine release **Cis atracurium-** -Better than atracurium -Metabolised by haff mann degeneration -No histamine release **Mivacurium** -Shortest acting -Smooth muscle relaxant of choice for day care surgery -Metabolized in plasma via pseudocholinesterase **Steroidal (Curoniums)** **Pancuronium** -2nd longest acting muscle relaxant -MR of choice in shock -CI in renal failure **Vecuronium** Most cardio-stable muscle relaxant	**Suxamethonium** -Fastest and shortest neuromuscular blocker -Metabolized in plasma via pseudocholinesterase -Deficiency can cause scholine apnea -can causes malignant hyperthermia

Rocuronium -painful muscle relaxant -use for rapid sequence intubation	

MALIGNANT HYPERTHERMIA **NEETPG 25,FMGE 25**

Pathophysiology- due to abnormal triggering of ryanodine receptor→ calcium released

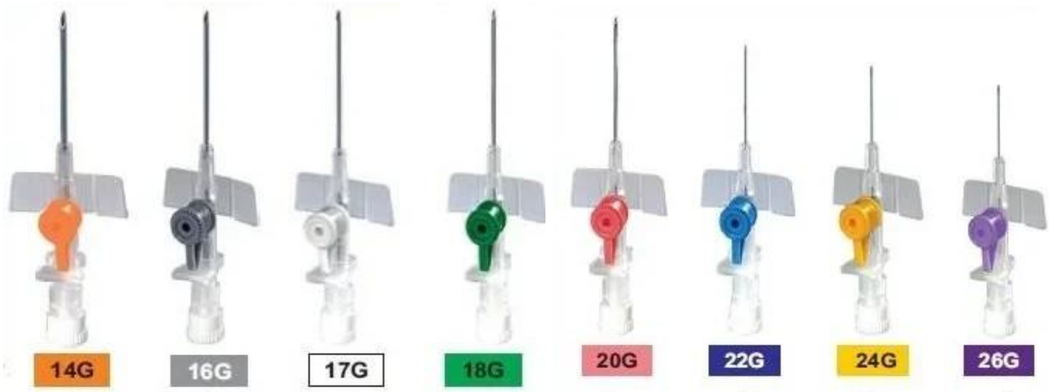
c/f- high temperature with sustained muscle contraction,tachycardia,DIC

Cause by- LA- lignocaine

- muscle relaxant- scholine
- Inhalational anesthesia- Halothane

Trt- dantrolene

IV CANNULA **FMGE 2020**



Gauge	Colour	Water flow (in mm)
14	Orange	270
16	Grey	180
17	White	125
18	Green	90
18	Green	90
20	Pink	60
22	Blue	36
24	Yellow	20
26	Violet	13

Foley's catheter size
G-green- 14
O-orange-16
R-red-18
Y-yellow-20
P-purple-22
B-blue-24

LARYNGEAL MASK AIRWAY **FMGE 2020,INI 24**

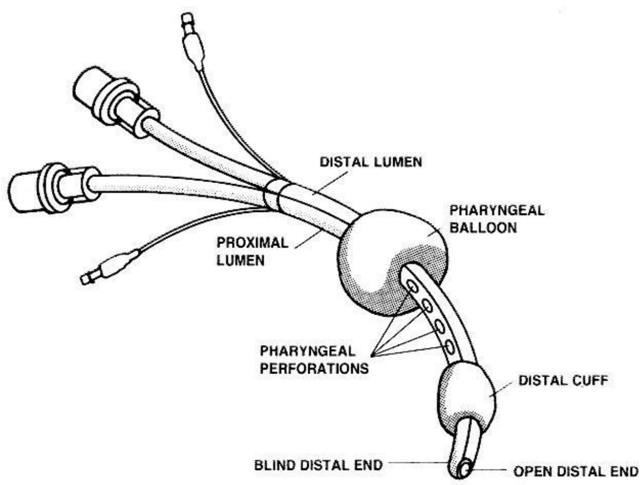
1. **Classic LMA-**1st generation

2. **Proseal LMA-** 2nd generation LMA

3. **Igel- 2nd generation**

4. **fastrack LMA-** 2nd generation LMA

TRACHESOPHAGEAL COMBITUBE

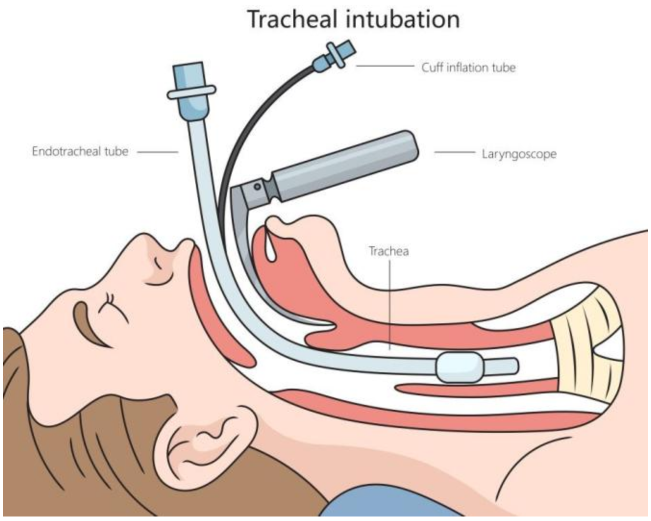


used in emergency situation

FMGE 2020

DOUBLE LUMEN TUBES- Indication: surgeries requiring one lung ventilation

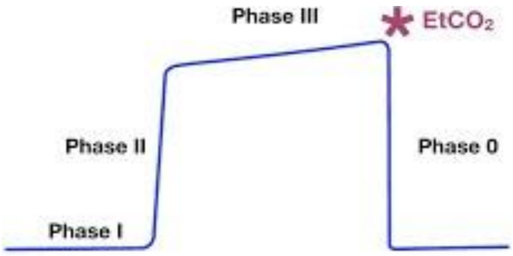
ENDOTRACHEAL INTUBATION- low pressure and high volume cuff used- ADULT
IOC to confirm- EtCo2



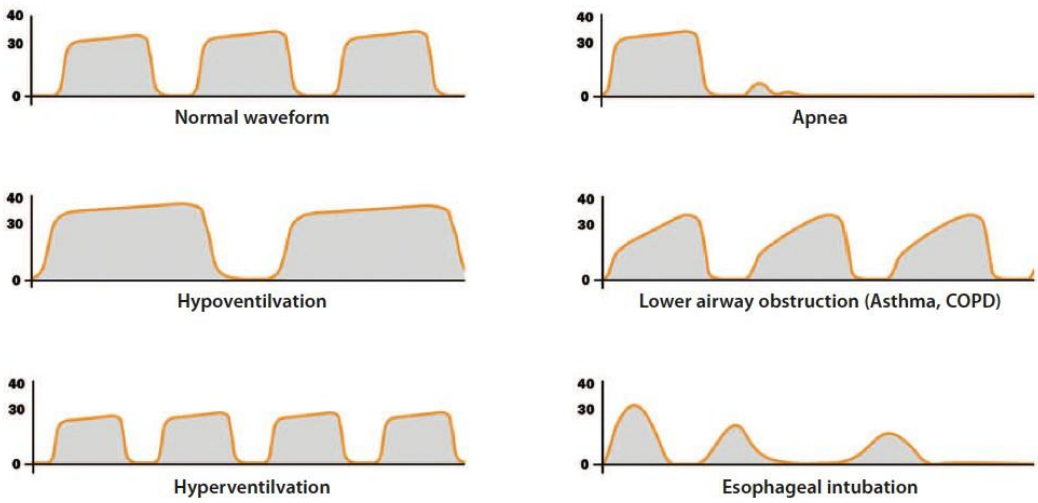
LARYNGOSCOPES

CAPNOGRAPHY **NEETPG 2020,21,24,INICET-21**

I- dead space ventilation
II- expiratory upstroke
III- alveolar plateau
0- inspiratory downstroke



The Normal Capnograph



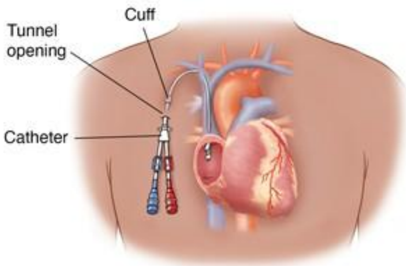
FOR CPR ASSESSMENT- CAPNOGRAPHY- keep etCO2 >20mmhg
VENOUS AIR EMBOLISM : entrapment of exogenous delivered gas or air into the venous vasculature . ETCO2 sudden drop .

SELLICKS MANEUVER **LARSON MANEUVER**

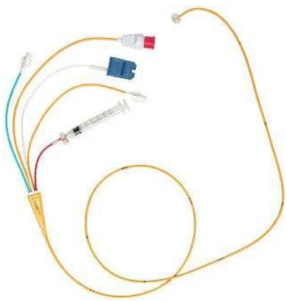
MC VEIN FOR CENTRAL LINE- IJV

MC VEIN FOR TPN- SUBCLAVIAN VEIN
MAXIMUM RISK OF INFECTION AND THROMBOSIS- FEMORAL VEIN

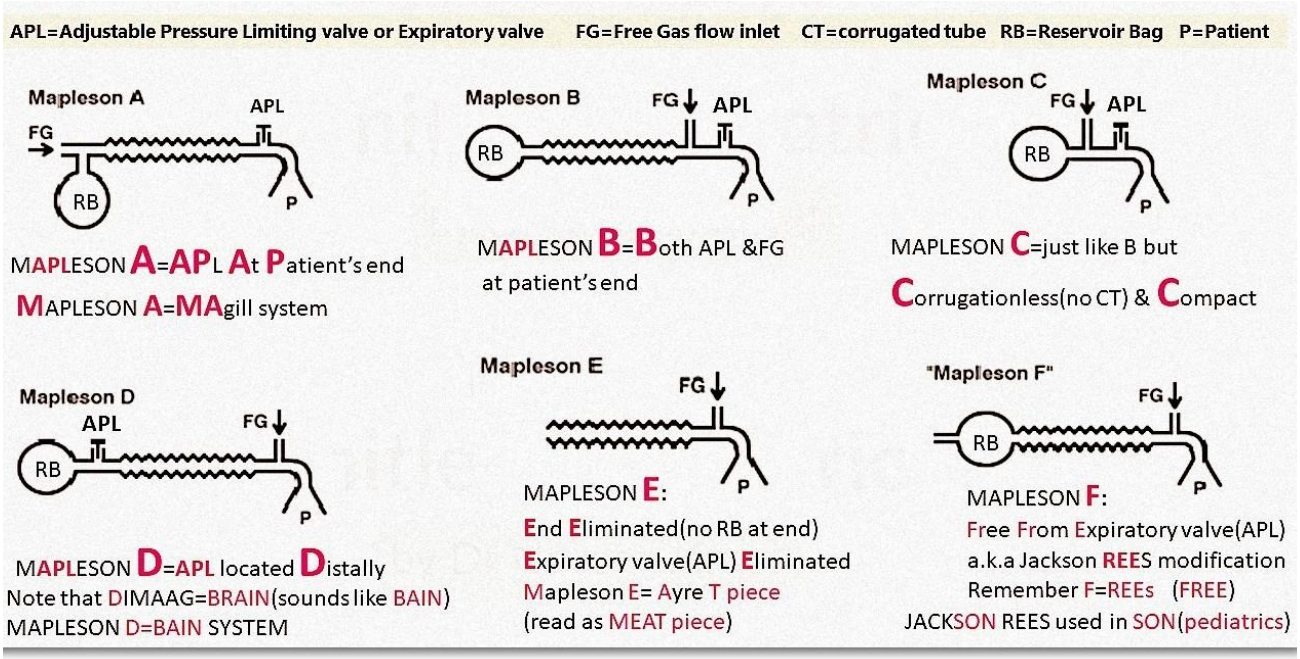

CHEMOPORT - CHEMOTHERAPY



Tunnel catheter- dialysis and TPN/CHEMO


swann ganz catheter- Pulmonary artery pressure

MAPELSON CIRCUIT **NEET PG 24**



Mapelson A-Best for **spontaneous respiration** (most efficient)
Worst for **controlled ventilation** (needs very high FGF)
Mapelson B- Best for **controlled ventilation** (most efficient)
Mapleson E (Ayre's T-piece)- used in children

BIS MONITORING

a processed EEG monitor reflecting the depth of anesthesia
40-60- general anesthesia (adequate for surgery)

CYLINDERS **INICET 2020,FMGE 2020,21**

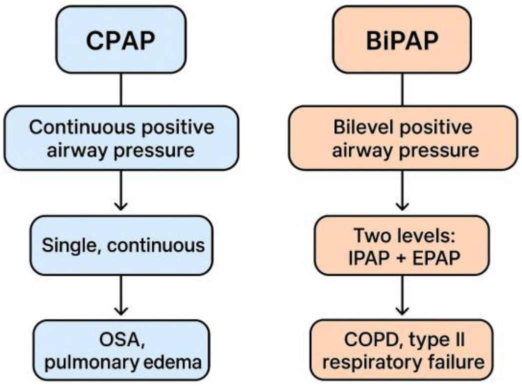
Cylinder	Color coding
O2	Black body white shoulder
Air	White body black shoulder
N2O	Blue body and shoulder
Entonoux (50% of N20 and 50% of 02) It provides analgesia and no anesthesia	Blue body and white shoulder
CO2	Grey

Helium	Brown
Cyclopropane	Orange

PIN INDEX

Gas	Pin Position
Oxygen (O_2)	2, 5
Nitrous Oxide (N_2O)	3, 5
Air	1, 5 & 2, 5 (dual pins)
Carbon Dioxide (CO_2)	1, 6
Helium	4, 6
Entonox (N_2O + O_2, 50:50)	7

MODES OF VENTILATION **FMGE 24**



MODE OF VENTILATION
CMV (controlled mandatory ventilation**)**→ Only machine, patient passive.
ACMV (assessed controlled mandatory ventilation**)** → Patient effort triggers → machine gives full breath (or machine gives backup if no effort).
SIMV (synchronised intermediate mandatory ventilation**)** → Mix of machine + patient → helps in weaning.

PEEP (Positive end expiratory pressure) **NEET PG 24 ,NEET PG 25**
prevent alveoli to collapse in end of expiration

USES OF PEEP
Improves oxygenation
Prevents atelectasis
↓ work of breathing

Used for ARDS, Pulmonary edema

Oxygen delivery devices **INICET 2020,21**

Devices	**Flow rate**	**Max saturation Fio2**
Nasal cannula	5 lit/min	40
Hudson mask	10 lit/min	60
venturi mask	15 lit/min	60
NRBM mask	15 lit/min	85
HFNC mask	60 lit/min	100

Venturi device : is preferred in copd patients

OPIOIDS
FENTANYL
Rapid onset
Side effect - Wooden chest syndrome

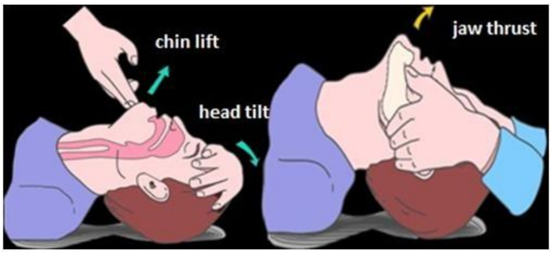
REMIFENTANIL
Shortest duration of action
Rapidly metabolized by non-specific esterase in tissue and plasma

CPCR **INICET 2020,23,24,NEET PG2023,24**
1. pulse +,breathing +,unconscious→ left recovery position
2. pulse+,breathing absent and unconscious→ rescue breath (1 breath in 6 sec)
3. pulse absent,breathing absent, unconscious→CPR immediate

CPR **NEETPG 2021**
Adult-
1 or >1 rescuer- 30:2
Depth- 5-6cm

Allow chest recoil
No. of compressions- 100-120/min

NEETPG-2021,22,23



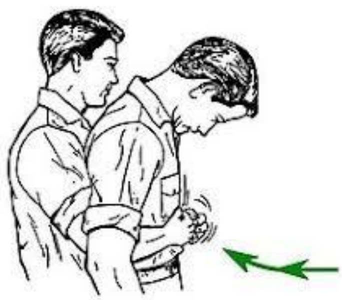
Infants-
depth 1.5 inches/4cm
No. of compressions- 100-120/min
2- thumb technique - >1 rescuer, 2- finger technique -1 rescuer
1 rescuer-30:2
2 rescuer-15:2
For neonates - Compression: Ventilation Ratio- 3:1

Adrenaline doses
Anaphylactic shock→ 1:1000 sc/im
Cardiac arrest→ 1:10,000 iv>io
Vasoconstriction- 1:1 lakh→ with LA for vasoconstriction- 1:2 lakh

IMP POINTS **NEETPG 2020,INI 2021**
Dobutamine stress test-most precise and prognostic for predicting risk of perioperative cardiac complications

Post operative nausea and vomiting - most frequent reason for hospital admission following day care surgery

Helmich manoeuvre - adult→ choking **FMGE 2021**



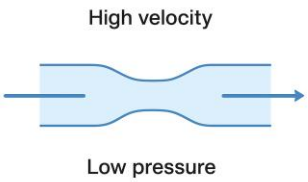
Neonate choking→ back slap, chest thrust
Neonate choking + unconscious→ chest compression **NEETPG 2025**

AHA- CHAIN OF SURVIVAL **FMGE 2025**

BERNOULLI PRINCIPLE **FMGE 2021**

Water flowing through a narrow pipe:
-In the narrow section → velocity ↑, pressure ↓.

Venturi effect (special case of Bernoulli principle)
-Used in Venturi masks → entrains air when oxygen flows at high velocity.



FMGE 2023

Airway resistance caused by an **endotracheal tube (ET tube)** is **mainly due to its internal diameter**.

Adu8 Bask LIfe Support Algorithm for Healthcare Providers

Verify scene safety.

- Check for responsiveness.
- Shout for nearby help.
- Activate ernerpencyresponse system via mobile device (if appropriate).
- Get AED and emergency equipment (or send someone to do so).

Normal breathing, pulse felt

Monitor until emergency responders arrive.

Look for no breathing or only gasping and check pulseiBlmultaneou ^ Is pulse **definitely** felt within 10 seconds?

No normal breathing, pulse felt

- Provide rescue breathing, 1 breath every 6 seconds or 10 breaths/min.
- Check pulse every 2 minutes; if no pulse, start CPR.
- If possible opioid overdose, administer naloxone if available per protocol.

flo breathing or only gasping. pulae notfeR

By this time in eli6cengrioa emergency response system or ba@up isa¢tivaleQ $ndAEDenBem$rgencyeQuIpmen1a*e nmmveoo spmeonei$netr|eMn8tñenn-

Start CPR

and 2 breaths.

- Use AEDessoonasltlsavaOabta.

AED arrives.

Chack rhy0xri.

shockable

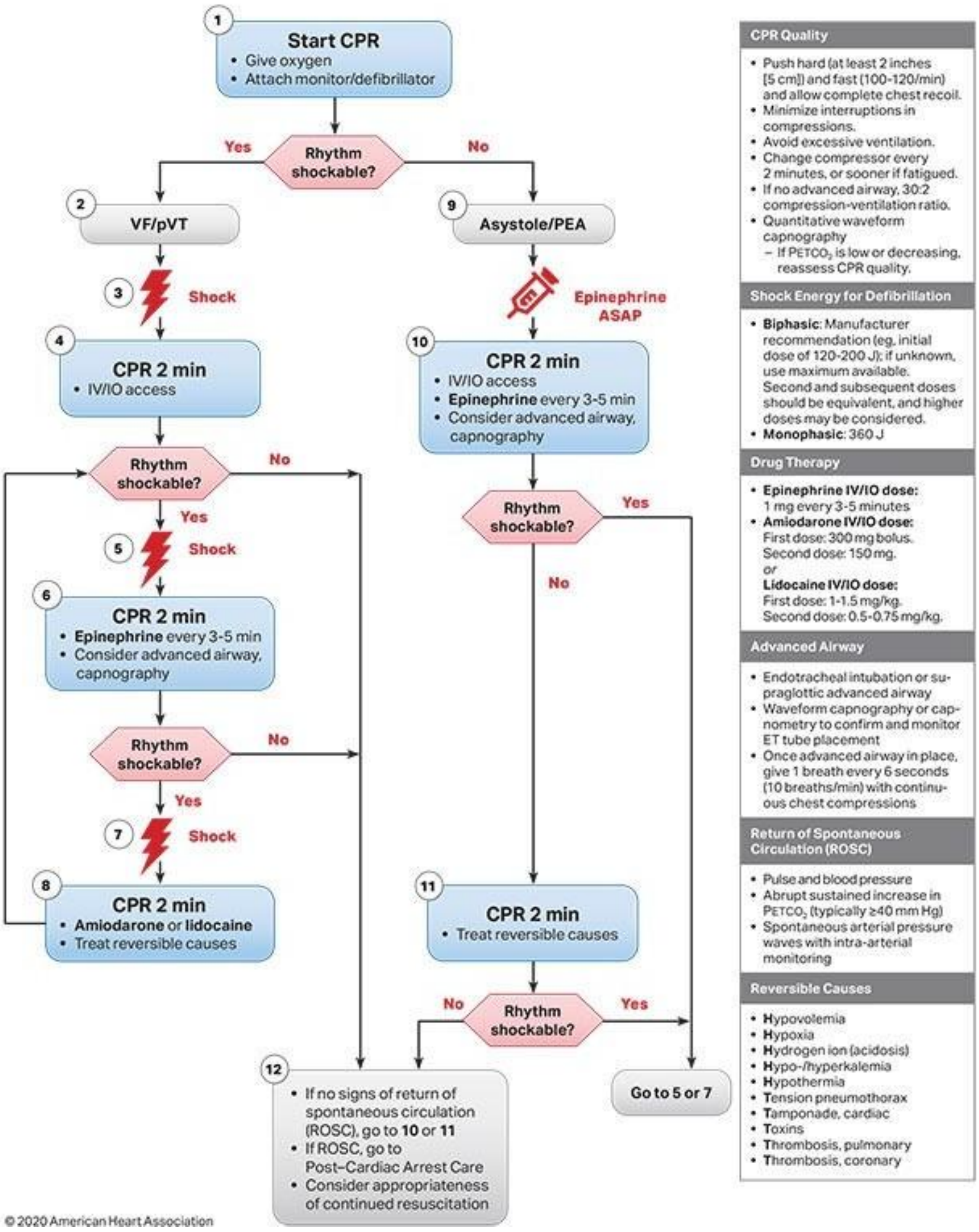
- 3lve shocl‹. Resume CPU fmrrwdtately for 2mlnutaB x t8 piomptad by AED toezoer rhythm check).

nancftockabla

- Resume CPR nv e0lalalytor 2 mlr¥J¢0s(unblpramptedby AEO \oalIt› rkyitxn chacl‹t
- Comi/xvunulALSpiovlderstake

NEETPG 2020

Adult Cardiac Arrest Algorithm (VF/pVT/Asystole/PEA)



CRICOTHYROIDECTOMY : NEET 2025
IT'S AN EMERGENCY AIRWAY MANAGEMENT .
Upper airway is by passed , resulting in less resistance to airway , work of breathing is less .
NEEDLE CRICOTHYROIDECTOMY : can be used only for 30-45 mins and
SURGICAL CRICOTHYROIDECTOMY : 20-48 HRS .
IT SHOULD BE FOLLOWED BY TRACHEOSTOMY.

Brachia; plexus block
Eg .. for shoulder dislocation interscalene block

NAME OF BLOCK	LEVEL OF BLOCK	ANALGESIA SEEN
Interscalene	Superior and middle trunk	Shoülder and upper arm. Most intense effect at C5-C7 dermatomes.
Supraclavicular	Distal trunk and proximal division	Elbow, forearm and hand
Infraclavicular	CORDS	Elbow, forearm and hand
Axillary	TERMINAL NERVES	FORE ARM AND HAND